

BUBBLE SORT

THE BUBBLE SORT EXPLAINED

- The bubble sort algorithm is considered to be one of the simplest sorting algorithms.
- Pairs of adjacent elements are compared and swapped if they are in the wrong order.
- This process is repeated until all elements in the list are sorted.
- Smaller, or larger elements 'bubble' to the top of the list.

BUBBLE SORT USES

- A bubble sort algorithm can be used:
 - mostly for educational purposes
 - in computer graphics
 - to sort an array of elements

EXAMPLE – 1ST PASS

13	4	24	10	3	33	28	52	6
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- The bubble sort algorithm compares 13 and 4, incorrect order, swap them

4	13	24	10	3	33	28	52	6
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- The algorithm then compares, 13 and 24, correct order.
- Compare 24 and 10, incorrect order, swap them.

4	13	10	24	3	33	28	52	6
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- Compare 24 and 3, incorrect order, swap them.

4	13	10	3	24	33	28	52	6
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- Compare 24 and 33, correct order.
- Compare 33 and 28, incorrect order, swap them.

4	13	10	3	24	28	33	52	6
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- Compare 33 and 52, correct order.
- Compare 52 and 6, incorrect order, swap them.

4	13	10	3	24	28	33	6	52
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EXAMPLE – 2ND PASS

4	13	10	3	24	28	33	6	52
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- Compare 4 and 13 correct order.
- Compare 13 and 10, incorrect order, swap them.

4	10	13	3	24	28	33	6	52
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- Compare 13 and 3, incorrect order, swap them.

4	10	3	13	24	28	33	6	52
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- Compare 13 and 24 correct order.
- Compare 24 and 28, correct order.
- Compare 28 and 33, correct order.

4	10	3	13	24	28	33	6	52
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- Compare 33 and 6, incorrect order, swap them.

4	10	3	13	24	28	6	33	52
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EXAMPLE – 3RD PASS

4	10	3	13	24	28	6	33	52
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- Compare 4 and 10, correct order.
- Compare 10 and 3, incorrect order, swap them.

4	3	10	13	24	28	6	33	52
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- Compare 10 and 13, correct order.
- Compare 13 and 24, correct order.
- Compare 24 and 28, correct order.
- Compare 28 and 6, incorrect order, swap them.

4	3	10	13	24	6	28	33	52
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EXAMPLE – 4TH PASS

4	3	10	13	24	6	28	33	52
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- Compare 4 and 3, incorrect order, swap them.

3	4	10	13	24	6	28	33	52
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- Compare 4 and 10, correct order.
- Compare 10 and 13, correct order.
- Compare 13 and 24, correct order.
- Compare 24 and 6, incorrect order, swap them.

3	4	10	13	6	24	28	33	52
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EXAMPLE – 5TH PASS

3	4	10	13	6	24	28	33	52
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- Compare 3 and 4, correct order.
- Compare 4 and 10, correct order.
- Compare 10 and 13, correct order.
- Compare 13 and 6, incorrect order, swap.

3	4	10	6	13	24	28	33	52
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EXAMPLE – 6TH PASS

3	4	10	6	13	24	28	33	52
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- Compare 3 and 4, correct order.
- Compare 10 and 6, correct order.

3	4	6	10	13	24	28	33	52
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And so on ...

BUBBLE SEARCH IMPLEMENTATION

```
public static void bubbleSort(int[] array){  
    int n = array.length;  
    int temp = 0;  
    for(int i=0; i < n; i++){  
        for(int j=1; j < (n-i); j++){  
            if(array[j-1] > array[j]){  
                temp = array[j-1];  
                array[j-1] = array[j];  
                array[j] = temp;  
            }  
        }  
    }  
}
```

BUBBLE SORT IMPLEMENTATION

- The bubbleSort method performs the bubble sort.
- One parameters – an array of integers.
- The variable n is defined as an integer and assigned the length of the array.
- The variable temp is declared as an int and initialised to 0.
- A loop traverses through the length of the array (1st pass, 2nd pass ...)
- A nested loop is used for the swapping process.
- Pairs of adjacent elements are compared.
- If the second element is greater than the first, the first is stored in temp, the second element is assigned to the first and temp is assigned to the second.

BIG O NOTATION – $O(N^2)$

- Given our example array where $n=9$:

13	4	24	10	3	33	28	52	6
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- The algorithm requires n iterations to perform the sort.
- Each iteration comprises a comparison and a swap if required.
- Given an array of size n (or 9 in our example), we need to perform $(n - 1) = 8$ comparisons on the first iteration.
- The second iteration requires $(n - 2) = 7$ comparisons and the third $(n - 3) = 6$
- This yields a formula as follows:

$$(n - 1) + (n - 2) + (n - 3) + \dots + 3 + 2 + 1 = \frac{n(n-1)}{2} = O(n^2)$$

- The time complexity of the bubble sort is simplified to $O(n^2)$ or $O(n^2)$ because as n grows, n^2 dominates the n term becomes inconsequential and therefore can be ignored.
- Our example where $n=9$ requires $8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 36$ swaps and notice $n * (n-1)/2$ can be evaluated as $(9 * 8)/2 = 36$